Homoeopathic Treatment in Hypothyroidism—An Open-Label, Prospective, Single-Arm, Clinical Trial

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Abstract

Background Hypothyroidism is one of the most common endocrinal disorders in India and its incidence is gradually increasing. A preliminary clinical trial was conducted to assess the probable role of individualised homoeopathic (IH) medicines in hypothyroidism with the assessment of hormonal titre (serum thyroid-stimulating hormone [TSH] and free T_4) and symptomatic improvement.

Methods An open-label, prospective, single-arm trial was conducted on 40 patients suffering from hypothyroidism at the outpatient of D. N. De Homeopathic Medical College and Hospital, West Bengal, India, and treated with IH. Serum TSH and FT4 were considered as the primary outcomes and Zuleswki's clinical scoring (ZCS) and thyroid symptom questionnaire (TSQ) as the secondary outcomes, measured at baseline and after 3 months of treatment. Intention-to-treat sample was analysed in the end by paired *t*-tests.

Results Total 58 patients were screened, of which 40 patients were enrolled; 4 patients dropped out, 36 patients completed the trial. Mean age was 44 years (standard deviation: 13.6), and majority of the patients were female (87.5%). Improvement in both primary and secondary outcomes was statistically significant ($p < 0.001$, paired *t*-tests), TSH (mean reduction: 2.7, 95% confidence interval [CI]: 1.9–3.6, $p = 0.001$), FT4 (mean increase: −0.2, 95% CI: −0.3 to −0.2, $p = 0.001$), ZCS (mean reduction: 3.7, 95% CI: 2.9–4.6, $p = 0.001$), and TSQ (mean reduction: 2.9, 95% CI: 2.1–3.6, $p = 0.001$). Most frequently used medicines were *Sulphur* (20%), *Pulsatilla nigricans* (12.5%), *Iodium* (10%), *Phosphorus* (10%), and *Natrum muriaticum* (7.5%).

Conclusion This study showed promising treatment effects of IH in hypothyroidism. Randomised trials are warranted (Trial registration: CTRI/2018/11/016451; UTN: U1111–1223–2292).

Keywords
- clinical trial
- homoeopathy
- hypothyroidism
- TSH
- free T_4

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Introduction

Hypothyroidism (ICD-10CM CodeE03.9)[1] refers to the common pathological condition of thyroid hormone deficiency. If untreated, it can lead to serious adverse health effects and ultimately death. Because of the large variation in clinical presentation and general absence of symptom specificity, the definition of hypothyroidism is predominantly biochemical. Overt or clinical primary hypothyroidism (OH) is defined as thyroid-stimulating hormone (TSH) concentrations above the reference range and free thyroxine concentrations below the reference range.[2] Secondary hypothyroidism can be differentiated in pituitary and hypothalamic by the use of thyrotrophin-releasing hormone (TRH) test.[3]

The prevalence of hypothyroidism in the developed world is ~4 to 5%[4,5] and that of subclinical hypothyroidism (SCH) is ~4 to 15%.[4,6] In India, hypothyroidism was usually categorised under the cluster of iodine-deficient disorders, typically assessed in school-aged children. Ever since India adopted the universal salt iodisation program in 1983, there has been a decline in goitre prevalence in several parts of the country, which were previously endemic.[7]

Clinical manifestations of hypothyroidism range from no signs or symptoms to life threatening events. The most common symptoms in adults are fatigue, lethargy, cold intolerance, weight gain, constipation, change in voice and dry skin, but clinical presentation can differ with age and sex, among other factors. The standard treatment is thyroid hormone replacement therapy with levothyroxine. However, a substantial proportion of patients who reach biochemical treatment targets have persistent complaints.[2] Homoeopathy though one of the most sought after complementary and alternative therapies, remained under-researched in such condition. An exploratory randomised control study was conducted by Chauhan et al in school children with SCH. The study concluded that homoeopathic intervention had not only the potential to treat SCH with or without anti thyroid autoantibodies (anti-TPOAb) but might also prevent progression to OH.[8] The aim of the present study was to assess changes in the serum TSH and freeT_4 (FT_4) in patients suffering from hypothyroidism.

Methods

Trial Design

This open label, prospective, non-randomised, non-controlled, single arm, pre-post comparison, observational trial was conducted among the patients of outpatient department and inpatient department of D. N. De Homoeopathic Medical College and Hospital, Kolkata, India. The study protocol was approved by the Institutional Ethical Committee (IEC) (Ref. No. DHC/Aca(PG)29/14/861/18) dated 13 September 2018 and was registered prospectively in the Clinical Trials Registry—India (Trial registration: CTRI/2018/11/016451; UTN: U1111–1223–2292). The full dissertation was submitted as postgraduate thesis of the corresponding author to the West Bengal University of Health Sciences.

Participants

Inclusion criteria were patients with newly diagnosed cases of hypothyroidism (serum TSH $\geq$ 4.70 µIU/mL), of either sex, aged between 15 and 65 years of age, of all socioeconomic conditions, with known but controlled systemic diseases and who were willing to give written consent to participate in the study. Patients undergoing levothyroxine sodium therapy included after discontinuation of the same followed by a washout period of 14 days if the patients agreed so; otherwise excluded. Exclusion criteria were patients already undergoing levothyroxine sodium therapy for hypothyroidism, TSH above 12 µIU/mL, patients suffering from congenital hypothyroidism, SCH, psychiatric diseases, suffering from any other uncontrolled systemic illness or life-threatening infection or any vital organ failure, already undergoing homoeopathic treatment for any chronic disease, substance abuse and/or dependence, self-reported immune-compromised state and women during pregnancy, puerperium and lactation and who were shifting from one system of medicine to other system of medicine frequently.

Intervention

Intervention was planned as administration of indicated homeopathic medicine in centesimal potencies based on totality of symptoms according to homoeopathic principles. Appropriate repetition was done at suitable intervals as per requirement. Each dose consisted of four cane sugar globules medicated with a single drop of the indicated medicine, preserved in 90% v/v ethanol. Each dose was directed to be taken orally on clean tongue with empty stomach. Duration of such therapy was 3 months. The patients were solely on homoeopathic treatment for the mentioned period of intervention. Medicines were obtained from Hahnemann Publishing Co. (HAPCO), SBL Pvt. Ltd. and Homoeopathy International—all good manufacturing practice certified firms. Single individualised medicine was prescribed on each occasion taking into account presenting symptom totality, clinical history details, constitutional features, miasmatic expressions, repertorisation using RADAR software when required with due consultation of Materia Medica. During follow-ups, either placebo or the same or other remedies in the same/higher potencies were given depending on the progress and nature of the cases. Medicine was selected on each occasion by the corresponding author (a postgraduate student) and the prescriptions were based on consultation with the Guide. In cases where there was any difference in opinion, it was resolved by involvement of another homoeopath who held masters' degree in Organon and Homoeopathic Philosophy with an experience of over 30 years. All the homoeopaths involved were affiliated with the respective state councils.

General Management

Every subject was advised adequate rest and balanced diet, to avoid goitrogenic food substances like cabbage, cauliflower, broccoli and soybean. Physical activity and exercise were encouraged and the subjects were advised to be present for monthly follow-ups.

Primary Outcome Measure

Patients under individualised homoeopathic (IH) treatment were assessed on the monitoring of the level of serum TSH and FT_4, before and after the treatment.

Secondary Outcome Measure

1. Zuleswki's clinical score (ZCS) for hypothyroidism: This simple scoring system is based on 7 subjective symptoms and 5 physical signs of hypothyroidism. Each symptom or sign if present gets 1 point and if absent gets 0 point. It's a 0 to 12 scoring scale, if total scoring >5 it defines hypothyroidism and if scores in between 0 and 2, it defines euthyroid status. This scale was originally written in English.[9]
2. Thyroid symptoms questionnaire (TSQ): TSQ is a simple questionnaire formed by British Thyroid Foundation Newsletter for assessing the well-being of the patient. Twelve simple questions are included here and give special emphasis on the mental symptoms along with common symptoms of hypothyroid. This questionnaire contains different kinds of questions like memory (questions 1–3), common symptoms of hypothyroid (questions 4–9), again literacy and nervousness (questions 10–12). It's a 4-point scale from better than usual to much less than usual 0, 0, 1, 1. Each question carries 1 point if present and 0 if absent.[10]

Sample Size

No relevant data were available on changes in serum TSH and FT_4or ZCS and TSQ scores by IH treatment using an open observational study design over 3 months of intervention. Thus, assuming a medium effect size (d) of 0.5, $\alpha = 0.05$ and power of 80%, to detect a significant intra-group changes, we would have required a sample size of 35. Keeping a provision for 15% drop-outs, target sample was 40.

Statistical Methods

The intention-to-treat (ITT) approach was followed, that is, every included patient entered the final analyses. Missing values were replaced by series mean values. Baseline descriptive data (categorical and continuous) presented in terms of absolute values, percentages, mean, standard deviations, confidence intervals (CI) etc. as appropriate. Parametric paired *t*-test was applied to compare the pre- and post-test data. Dependent observations of continuous outcomes at baseline and at different points of time were compared using paired *t*-test. Effect sizes (Cohen's ***d***) were reported 3 months after treatment. *p*-Values were set at less than 0.05 two-tailed as statistically significant.

Results

Participant Flow

According to pre-specified inclusion and exclusion criteria, 58 patients suffering from hypothyroidism were screened; out of which 18 were excluded on account of varied reasons and remaining 40 patients fulfilled the eligible criteria included and were ultimately enrolled. Baseline socio-demographic and outcome data was obtained for these 40 enrolled patients and were subjected to IH treatment. Before participation in the trial, five patients were on levothyroxine sodium therapy. Rest were either reluctant to take usual care or were not sure about their hormonal status. After 3 months of intervention, outcome data was recorded again. During course of treatment, 4 patients were dropped out, while 36 patients completed the entire period of trial (►**Fig. 1**).

Recruitment

Starting from December 2018, follow-up of the last enrolled patient was completed by the end of March 2020.

Baseline Data

Eleven features were studied—age, gender, duration of suffering, co-morbidities, food habits, body mass index, residence, marital status, educational status, employment status and family income status (►**Table 1**).

Numbers Analysed

Outcomes from 36 patients were completed; however, all the enrolled subjects ($n = 40$) entered into the final analyses.

Data Distribution

Paired *t*-test was performed to analyse the statistical significance of assessment of serum TSH, FT_4, ZCS and TSQ scoring in cases of hypothyroidism under the IH treatment.

Outcomes and Estimation

Improvement in both primary and secondary outcomes was statistically significant (p <0.001, paired *t*-tests). Intra-group changes over 3 months were statistically significant in all four outcome measures ($p < 0.05$), TSH (mean reduction: 2.7, 95% CI: 1.9–3.6, $p = 0.001$), FT4 (mean increase: −0.2, 95% CI: −0.3 to −0.2, $p = 0.001$), ZCS (mean reduction: 3.7, 95% CI:

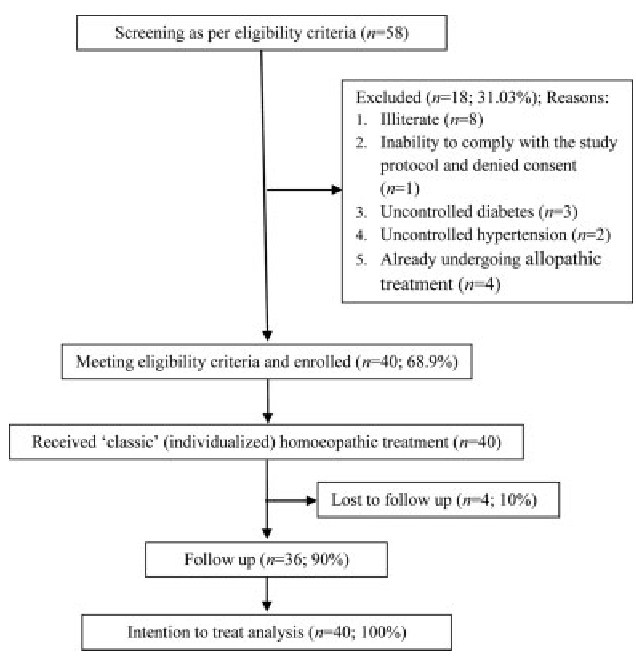

Fig. 1 Study flow diagram.

Table 1 Socio-demographic characteristics of the patients at baseline ($n = 40$)

Features	*n*	%
Age (y) [a]	44.0 ± 13.6	–
Sex [b]		
■ Male	5	12.5
■ Female	35	87.5
Duration of suffering (mo) [a]	4.3 ± 2.9	–
Co-morbidities [b]		
■ Rheumatism	8	20
■ Insomnia	3	7.5
■ Migraine	3	7.5
■ Urinary tract infection	3	7.5
■ Alopecia	2	5
■ Dermatitis	2	5
■ Dysmenorrhoea	2	5
■ Menorrhagia	2	5
■ Obesity	2	5
■ Gastritis	1	2.5
■ Hoarseness	1	2.5
Food habit [b]		
■ Veg	8	20
■ Non-veg	32	80
Body mass index [a]	26.4 ± 2.7	–
Blood pressure (mm of Hg) [a]		
■ Systolic	120.3 ± 8.6	–
■ Diastolic	80.1 ± 5.0	–
Residence [b]		
■ Rural	14	35
■ Urban	26	65
Marital status		
■ Single	3	7.5
■ Married	37	92.5
Educational status [b]		
■ 10^{th} std. or less	13	32.5
■ Higher than 10^{th} std.	9	22.5
■ Graduate or above	18	45
Employment status [b]		
■ Business	3	7.5
■ Service	3	7.5
■ Dependent	34	85
Family income status [b]		
■ Poor	5	12.5
■ Middle	19	47.5
■ Affluent	16	40

[a]Continuous data presented as means ± standard deviations.
[b]Categorical data presented as absolute values (percentages).

2.9 to 4.6, $p = 0.001$) and TSQ (mean reduction: 2.9, 95% CI: 2.1–3.6, $p = 0.001$; ►**Table 2**).

Medicines Used

Sixteen medicines were prescribed at baseline. Among them *Sulphur* ($n = 8$, 20%), *Pulsatilla nigricans* ($n = 5$, 12.5%), *Iodium* ($n = 4$, 10%), *Phosphorus* ($n = 4$, 10%) and *Natrum muriaticum* ($n = 3$, 7.5%) were the most frequently prescribed medicines, followed by *Calcarea carbonicum, Graphites, Medorrhinum, Thuja occidentalis, Thyroidinum, Kali carbonicum, Lachesis mutus, Lapis alba, Lycopodium clavatum, Natrum sulphuricum,* and *Tuberculinum bovinum.* Percentages were calculated by dividing the number of prescriptions of each medicine by total number of prescriptions (i.e., 40) multiplied with 100 (►**Table 3**). Indications of the prescribed medicines are given in ►**Table 4**. The symptoms which were clinically verified were mentioned in the table.

Adverse Events

Patients were instructed to report any harms, unintended effects, serious adverse events and undue aggravations either directly in the outpatients or over phone during the study period. No serious adverse event was noted during the trial. Two cases presented with mild indisposition that was managed with dietetic modification.

Discussion

Principal Findings

An open label, prospective, single arm, pre-post comparison trial was conducted at D. N. De Homoeopathic Medical College and Hospital among 40 patients suffering from hypothyroidism and was treated with indicated IH medicines following the homoeopathic principles. Level of serum TSH, FT_4 and ZCS, TSQ scores were taken as primary and secondary outcome measures respectively. Statistically significant reduction in serum TSH and FT_4 levels as well as improvement were seen in ZCS and TSQ scoring. Intra-group changes over 3 months were statistically significant in all primary and secondary outcome measures. Twenty-six patients showed disappearance of symptoms following Kent's observations and Hering's law. Most commonly used medicines were *Sulphur*, *Pulsatilla nigricans*, *Iodium*, *Phosphorus* and *Natrum muriaticum*.

Strengths of the Study

It was an open level, prospective, pre-post comparison trial. Primarily screening was done by Billewicz diagnostics index for hypothyroidism; scoring ≥ 25 was considered for primary screening, so the chances of bias were less. The methodological strengths of the study include the use of standardised and already validated outcome scales—severity of disease measuring by hormonal level along with ZCS for hypothyroidism and TSQ score. Enrolment into the study was prospective, that is, the protocol was declared before the enrolment of the first patient.

The study was clear in terms of declaration of protocol, ethical conduct and reporting. There was no placebo control and no new drug was experimented; thus, ethically less vulnerable.

Table 2 Comparison of serum TSH, free T4, Zuleswki's clinical scoring and thyroid symptom questionnaire at baseline and over 3 months ($n = 40$)

Outcome measures	Baseline: mean ± SD	After 3 months: mean ± SD	Mean difference (SE)	95% CI	t_{38}	p-Value
Serum TSH (μIU/mL)	7.7 ± 1.5	5.0 ± 2.4	2.7 (0.4)	1.9, 3.6	6.767	0.001[a]
Free T4 (ng/dL)	0.7 ± 0.1	0.9 ± 0.2	−0.2 (0.0)	−0.3, −0.2	−6.587	0.001[a]
Zuleswki's clinical scoring	7.3 ± 1.5	3.5 ± 2.7	3.7 (0.4)	2.9, 4.6	9.021	0.001[a]
Thyroid symptom questionnaire	7.3 ± 1.1	4.4 ± 2.0	2.9 (0.3)	2.1, 3.6	8.186	0.001[a]

Abbreviations: CI, confidence interval; SD, standard deviation; SE, standard error; t_{38}, t score at 38 degrees of freedom; TSH, thyroid-stimulating hormone.
p-Value reflects intra-group differences detected by paired t-tests.
[a]$p < 0.05$ considered as statistically significant.

Table 3 List of the medicines prescribed at baseline ($n = 40$)

Sl. no.	Name of the medicines	n	%
1.	*Sulphur*	8	20
2.	*Pulsatilla nigricans*	5	12.5
3.	*Iodium*	4	10
4.	*Phosphorus*	4	10
5.	*Natrum muriaticum*	3	7.5
6.	*Calcarea carbonicum*	2	5
7.	*Graphites*	2	5
8.	*Medorrhinum*	2	5
9.	*Thuja occidentalis*	2	5
10.	*Thyroidinum*	2	5
11.	*Kali carbonicum*	1	2.5
12.	*Lachesis mutus*	1	2.5
13.	*Lapis alba*	1	2.5
14.	*Lycopodium clavatum*	1	2.5
15.	*Natrum sulphuricum*	1	2.5
16.	*Tuberculinum bovinum*	1	2.5

Before enrolling of each patient, they were provided with patient information sheet in local vernacular Bengali detailing the study aims and objectives, methods, risks and benefits of participating and confidentiality issues. After which, written informed consent was obtained. Thus, the study conformed to every possible ethical standard. Clearance was obtained from the IEC. Every patient was treated with IH medicine based on totality of symptoms and homoeopathic principles and all the collected data (hard form) was converted into an analysable and reproducible master chart (soft copy) where all data was extracted systematically and underwent statistical analysis subsequently. Though the sample size of this study was small, it was still adequately powered to detect changes in the specified outcome measure over 3 months. Medicines were prescribed based on the characteristic symptoms and the clinically verified symptoms were mentioned against each indicated medicine (►**Table 4**).

Weakness of the Study

Duration of follow-up was one of the major limitations of the study. It could have been enhanced and/or the outcomes could have been measured repeatedly to enhance the rigor of the study. This study did not support conclusions as to the efficacy or effectiveness of the homoeopathic medicines because no methodology for this purpose (control group, randomisation, blinding, multi-centre study design etc.) was built into its design.

Another limitation was inherent in the observational design of the study, that is, no causal inference could be attributed to the treatment and the treatment could not be regarded as the sole reason for the outcome changes in the patients. In this study, homoeopathic constitutional anti-miasmatic medicines were mainly used. But action of short-acting palliative medicines remained obscure. The study included only a small group of patients. So, greater number of patients from different demographic regions might have given better result. Since large within-individual variations exist in serum TSH and FT4 and the laboratory reference ranges are relatively insensitive to these variations, a single reading of serum TSH and FT4 level is not sufficient for diagnosis and conclusion. This should be treated as one of the study limitations and needs to be addressed in future trials by taking multiple readings.

Strength and Weakness in Relation to Other Studies

A report on patient-centred management of hypothyroidism by Kalra et al focused on means of helping persons with hypothyroidism to live a healthy life, through education about precaution in concomitant food and medications intake, as well as sick day management.[11] An observational cross-sectional study by Kumar et al concluded that public health measures were required to improve knowledge, awareness and practices in patients with primary hypothyroidism.[12] There have been many articles published in different journals regarding homoeopathic treatment on hypothyroidism. Different kinds of approach were taken like efficacy of some fixed medicine on hypothyroidism, some of them were randomised control trial (RCT), observational trial, some of them were conducted only on females, some assessed the efficacy of using particular scale etc. All of them have been found in favour of homoeopathic treatment. In a prospective randomised trial conducted on 10 patients between 20 to 40 years of age suffering from primary hypothyroidism showed that

Table 4 Indications of the prescribed medicines with clinical verification

Medicines used	Physical generals	Mind symptoms	Clinically verified symptoms
Sulphur	Swelling and inflammation of neck History of suppressed skin eruptions Desire for sweets, salts Dislike for bathing Burning sensation in palms, soles Offensive stool and urine	Mental irritability Forgetfulness Egoistic	Mental irritability Swelling and inflammation of neck Burning sensation in palms, soles Offensive stool and urine
Pulsatilla nigricans	Swelling of glands of neck General sensation of chilliness < evening, prefers open air Irregular menstruation, delayed menses, chilly patient, thirstlessness	Desire company, mild gentle calm quiet patient Weeping disposition	Swelling of glands of neck Irregular menstruation Thirstlessness
Iodium	Great weakness and prostration, swelling and induration of glands, thirst for large quantities of cold water, ravenous hunger	Lachrymose disposition and mental dejections Cross, irascible, peevish	Lachrymose disposition Great weakness and prostration Swelling and induration of glands
Phosphorus	Swelling of neck. Oversensitivity to light, noise, sound Craving for cold things, ice cream, cold water, salt	Sympathetic, indifferent, timid, irresolute, fearful, apathetic unwilling to talk	Indifferent, fearful Oversensitivity to light, noise, sound
Natrum muriaticum	Throat and neck emaciate rapidly Goitre of a large size Takes extra salt during eating Sun-heat intolerance Lean, thin, emaciated body, sleep disturbed	Weeping disposition, catches cold easily, mental grief, dreams of dead person	Mental grief Dreams of dead person Goitre of a large size Takes extra salt during eating Sleep disturbed
Calcarea carbonicum	Hard and strumous swelling of the thyroid gland Fair, fatty, flabby, chilly and constipated, generally feels better when constipated Desire for sweet, egg Cold hands and feet, tendency to catch cold easily, sour eructation	Apprehensive forgetfulness	Forgetfulness, Hard and strumous swelling of the thyroid gland Constipation Sour eructation
Graphites	Swelling of the glands of the neck Chilly patient, obese, constipated patient, skin affection on the fold of skin, aversion to meat, catches cold easily, general tendency to suppuration	Sad, gloomy, melancholic patient	Swelling of the glands of the neck Constipation Skin affection on the fold of skin
Medorrhinum	Spasms of neck muscles Ravenous hunger, excessive thirst even dreams of drinking, sleeps in knee chest position, desire for sweet, salt, leucorrhoea fishy odour	Weak memory, forgetfulness, weeping disposition	Spasms of neck muscles. Leucorrhoea fishy odour
Thuja occidentalis	Swelling of glands of neck Recurrent urinary tract infections. Breakfast diarrhoea, tea drinking dyspepsia Sweat only covered part Dislikes meat	Suspicious, obstinate Religious	Obstinate Swelling of glands of neck Recurrent urinary tract infections Tea drinking dyspepsia
Thyroidinum	Thyroid very slightly enlarged Swelling of glands of stony hardness Rheumatic arthritis with tendency to obesity, irritable from least contradiction, sensitive to cold, easy fatigue, desire sweet, great thirst	Irritable worse least opposition Depression Delirium of persecution	Irritable worse least opposition Depression Thyroid enlargement with stony hardness Great thirst
Kali carbonicum	Goitre—weakness of muscles of neck Menses too early, profuse with severe backache specially very early morning, chilly, obese patient, with weakness and profuse sweat	Sadness with tears Peevish humour Vexed and irritated mood	Sadness with tears Goitre Menstruation with moderate flow
Lachesis mutus	Neck excessively sensitive to least pressure Insomnia, very hot patient Intolerance to sun's-heat Wears clothes loosely Unrefreshing sleep	Loquacious, suspicious, forgetful	Forgetful Unrefreshing sleep
Lapis alba	Induration and slight swelling of thyroid gland Fatty, anaemic, ravenous appetite, h/o otitis media	–	Induration and slight swelling of thyroid gland
Lycopodium clavatum	Swelling of glands of neck and of shoulder For persons intellectually keen, but physically weak	Silent, melancholy and peevish humour Obstinacy, mistrustful	Obstinacy Swelling of glands of neck

Table 4 (*Continued*)

Medicines used	Physical generals	Mind symptoms	Clinically verified symptoms
	Deep seated, progressive, chronic diseases Affects right side of the body Desire for sweet things		
Natrum sulphuricum	Stitches in nape of neck at night History of head injury Tongue: thick brown coating Leucorrhoea: yellowish	Sadness, inclined to weep Suicidal tendency	Sadness, inclined to weep Leucorrhoea: yellowish
Tuberculinum bovinum	Glands in neck swollen and very tender Menses too early, too profuse, with pain during periods Chilly patients, catches cold easily. Aversion to meat Early morning diarrhoea	Anxiety, gloomy, melancholy humour. Fear of dogs, animals.	Anxiety, gloomy, melancholy Menses too early, too profuse, with pain during periods Early morning diarrhoea

Thyroidinum 3x was considered as a specific supplement for hypothyroidism and it had a curative effect on hypothyroidism without any adverse effects along with it helped in reduction in weight.[13] Another study concluded that *Thyroidinum* 3X had effect on anti-TPO Ab titres and prevented progression to OH.[14] An exploratory RCT on SCH with or without autoimmune thyroiditis on 194 children (6–18 years) showed homoeopathic intervention could reduce TSH and anti-TPOAb titre (70.29%) along with it might also prevent progression to OH (10.5%).[8] An extramural research was conducted on post-iodinisation scenario of school children in Delhi which suggested that environmental goitrogens might be responsible for goitre prevalence in the region.[15] Another randomised, placebo-controlled, double-blind clinical trial suggested that by predefining the primary outcome measure in a different way, an augmented reduction in weight on day one after treatment with homoeopathic medicine *Thyroidinum* 30cH might be demonstrated.[16] A study on animal model by Guedes et al suggested that the number of frogs *Rana catesbeiana* that reached the four-legged stage during metamorphosis was generally smaller in the treated group consisting of high-diluted solution of thyroid glands, than in the hydroalcoholic control group. It was postulated that thyroid hormones transmitted information specific to the molecules used to prepare the solution, even though the molarity was beyond Avogadro's number.[17]

Unanswered Questions and Future Research

The present study was an observational trial so randomised controlled design would have been kept simultaneously to verify the effectiveness of treatment. Large number of patients from different geographical areas with extended time of research would provide better results.

In this study, most of the polychrest medicines were used, so short-acting palliative medicines and their actions on hypothyroid conditions remained unknown. So, these drugs might have been taken into consideration.

Conclusion

There was statistically significant reduction in serum TSH and FT_4 levels as well as improvement was seen in ZCS and TSQ scoring in cases of hypothyroidism after 3 months of IH treatment. Further exploration by randomised trials with enhanced methodological rigor are warranted.

Authors' Contribution

ASR, SG, SKM, PH, and SD were involved in concept, literature search, clinical study, data acquisition, data interpretation, and preparation of the article; AN, MK, and SS were involved in concept, literature search, design, data interpretation, statistical analysis, and preparation of the article. All the authors reviewed and approved the final paper.

Ethical Approval

The study protocol was approved by the IEC (Ref. No. DHC/Aca(PG)29/14/861/18) dated 13 September 2018 and was registered prospectively in the Clinical Trials Registry—India (Trial registration: CTRI/2018/11/016451; UTN: U1111–1223–2292).

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Nil.

Conflict of Interest

We declare no conflict of interest. The study was conducted as postgraduate thesis of the corresponding author.

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References

1 Human Diseases in ICD-11 Classification by World Health Organization. [Accessed September 20, 2020]. Available from: https://www.icd10data.com/ICD10CM/Codes/E00-E89/E00-E07/E03-/E03.9
2 Chaker L, Bianco AC, Jonklaas J, Peeters RP. Hypothyroidism. Lancet 2017;390(10101):1550–1562
3 Kostoglou-Athanassiou I, Ntalles K. Hypothyroidism - new aspects of an old disease. Hippokratia 2010;14(02):82–87
4 Hollowell JG, Staehling NW, Flanders WD, et al. Serum TSH, T(4), and thyroid antibodies in the United States population (1988 to

1994): National Health and Nutrition Examination Survey (NHANES III). J Clin Endocrinol Metab 2002;87(02):489–499

5 Hoogendoorn EH, Hermus AR, de Vegt F, et al. Thyroid function and prevalence of anti-thyroperoxidase antibodies in a population with borderline sufficient iodine intake: influences of age and sex. Clin Chem 2006;52(01):104–111

6 Bemben DA, Hamm RM, Morgan L, Winn P, Davis A, Barton E. Thyroid disease in the elderly. Part 2. Predictability of subclinical hypothyroidism. J Fam Pract 1994;38(06):583–588

7 Unnikrishnan AG, Kalra S, Sahay RK, Bantwal G, John M, Tewari N. Prevalence of hypothyroidism in adults: an epidemiological study in eight cities of India. Indian J Endocrinol Metab 2013;17(04):647–652

8 Chauhan VK, Manchanda RK, Narang A, et al. Efficacy of homeopathic intervention in subclinical hypothyroidism with or without autoimmune thyroiditis in children: an exploratory randomized control study. Homeopathy 2014;103(04):224–231

9 Zulewski H, Müller B, Exer P, Miserez AR, Staub JJ. Estimation of tissue hypothyroidism by a new clinical score: evaluation of patients with various grades of hypothyroidism and controls. J Clin Endocrinol Metab 1997;82(03):771–776

10 Saravanan P, Chau WF, Roberts N, Vedhara K, Greenwood R, Dayan CM. Psychological well-being in patients on ‘adequate’ doses of l-thyroxine: results of a large, controlled community-based questionnaire study. Clin Endocrinol (Oxf) 2002;57(05):577–585

11 Kalra S, Agarwal N, Aggarwal R, et al. Patient-centered management of hypothyroidism. Indian J Endocrinol Metab 2017;21(03):475–477

12 Kumar P, Khandelwal D, Mittal S, et al. Knowledge, awareness, practices and adherence to treatment of patients with primary hypothyroidism in Delhi. Indian J Endocrinol Metab 2017;21(03):429–433

13 Sangita A, Jadav SB. Efficacy of homoeopathic thyroidinum 3x in hypothyroidism in both sexes between age group 20 to 40 years. Int J Sci Res 2019;8:800–809

14 Ghare P, Jadhav AB, Patil AV. A clinical study to see the effect of thyroidinum, a homoeopathic preparation on thyroid peroxidase antibody in subclinical hypothyroidism of age group between 18–70 years. Int J Health Sci Res 2020;10:18–22

15 Chauhan VK, Manchanda RK, Narang A, et al. Prevalence of thyroid disorders in school children in Delhi – post iodization scenario. Indian J Res Homoeopathy 2009;3:29–33

16 Schmidt JM, Ostermayr B. Does a homeopathic ultramolecular dilution of Thyroidinum 30cH affect the rate of body weight reduction in fasting patients? A randomised placebo-controlled double-blind clinical trial. Homeopathy 2002;91(04):197–206

17 Guedes JRP, Ferreira CM, Guimarães HMB, Saldiva PH, Capelozzi VL. Homeopathically prepared dilution of Rana catesbeiana thyroid glands modifies its rate of metamorphosis. Homeopathy 2004;93(03):132–137